

2025-06-11 NSF S&CC – CommonSenses project – sensor box data specifications

Multiple types of messages pushed up through the Particle interface

Type #1: data from a sensor box

We collect temperature (T), relative humidity (RH), and noise sensor readings every minute. However data are only pushed up to the Particle.io system every 15 minutes (this saves power in booting up the LTE module only infrequently rather than keeping it open).

This means one “message” will have 15 T, RH, and Noise data readings, each for a consecutive minute.

The message starts with the timestamp ('month', 'day', 'hour', and 'minute') as **UTC time stamp** of the first sensor reading in the list). For each subsequent reading we should add 1 minute to the time stamp.

The sensor data are also uploaded as integers that need post-processing to get to the units we want to report in, also to minimize data transmission (power consumption).

Generic format of a message:

,BoxID,month,day,hour,minute,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,
Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,
.RH,Noise,T,RH,Noise

Example:

,19,6,6,17,34,2483,5480,6389,2482,5483,5181,2477,5481,5222,2484,5486,6121,2485,5487,5649,
2483,5481,6591,2483,5482,6923,2488,5484,6904,2484,5482,6860,2480,5482,6731,2487,5488,68
29,2487,5491,6762,2485,5486,6515,2479,5488,6711,2488,5489,6806

In the example:

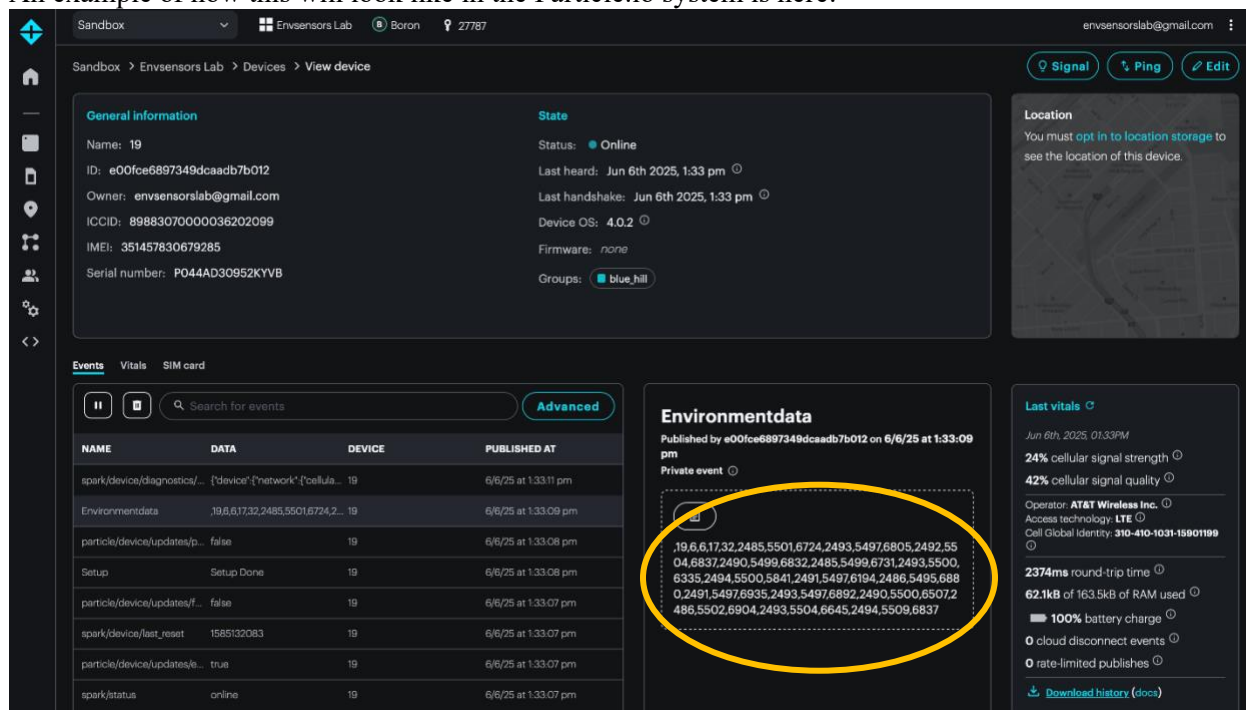
- Box ID is 19
- Time stamp (of first sample) is 6/6/2025 17:34 UTC time – i.e., the first T, RH, and Noise data were collected at 17:34 UTC time, and the second T, RH, and Noise data were collected at 17:35 UTC time, etc.
- The first “batch” of data are: T_{raw} = 2483, RH_{raw} = 5480, Noise_{raw} = 6389

Postprocessing needed:

- Temperature should be divided by 100 to get to the actual reading **in degrees C**.
 - In the example above $T_raw = 2483 \rightarrow T_actual = \underline{24.83\text{ C}}$
 - The sensor only has accuracy around 0.1 C at best, so this can be rounded to keep just one decimal of precision in our database.
- Relative humidity should be divided by 100 to get to the actual reading **in %**.
 - In the example above $RH_raw = 5480 \rightarrow RH_actual = \underline{54.80\%}$
 - The sensor only has accuracy around 1 % (RH is kind of tough to measure), so we can choose to round or keep one decimal of precision in our database.
- Noise is a count-based reading from a 16-bit analog-to-digital converter (signed integer) and must be converted to **dB** using the following equation: **dB value = (Noise_raw / 32768) * 6.144 * 50**
 - In the example above $Noise_raw = 6389 \rightarrow Noise_actual = (6389 / 32768) * 6.144 * 50 = \underline{59.90\text{ dB}}$
 - Depending how we want to treat analyses, we might want to keep 1 digit of precision, but for displaying people probably don't need to see anything other than the integer part of the number.

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An example of how this will look like in the Particle.io system is here:



Type #2: error reported from a sensor box

Errors that happen in the code which will indicate a hardware malfunction are also reported. These would ideally be reported via email or similar to the research team, as it would indicate some physical action (probably visiting the sensor) is needed.

Generic format of a message:

BoxIDplusDate Hour:Minute ErrorCode

where *BoxIDplusDate* is 2-digit BoxID followed by 2-digit year, 2-digit month, and 2-digit date

Example (first highlighted message in screenshot below):

19250610 14:42 E10

In the example:

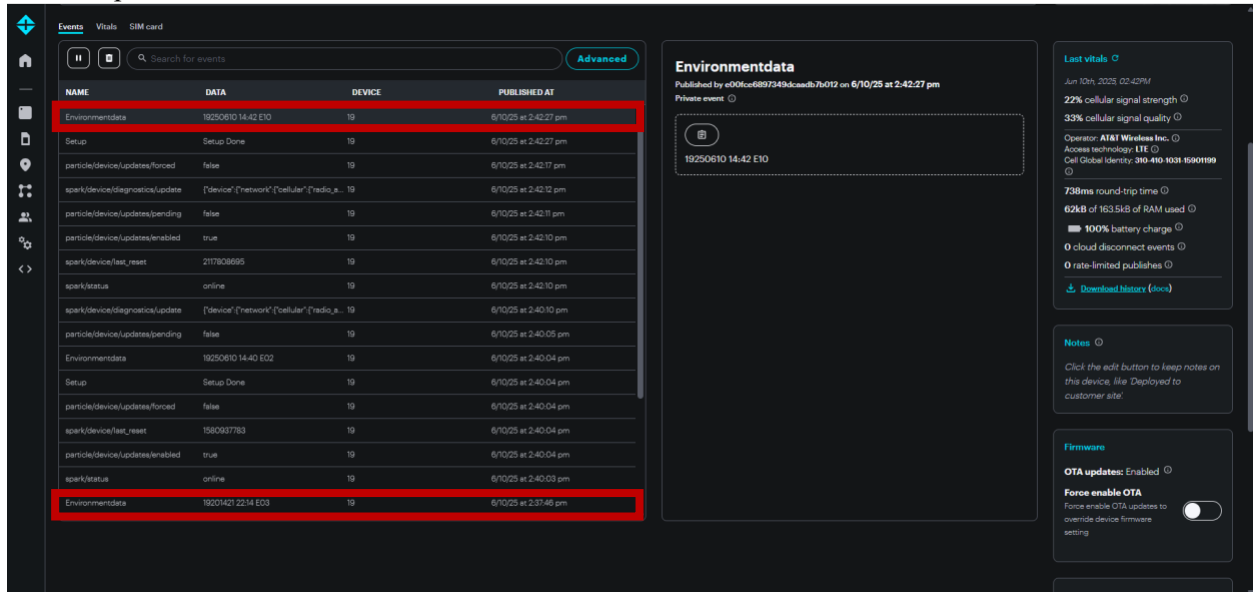
- Box ID is 19
- Date and time: 2025-06-10 14:42 UTC
- Error number is 10 (from table below this means “ADC not found”)

Error Code	Error Description
E1	Serial Port not open
E2	T&RH sensor not found
E3	RTC not found
E4	LTE module connection to cloud timeout (only log to SD card)
E5	LTE module receiving data from Arduino timeout (only log to SD card)

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E6	SD module failure
E7	Buffer overflow
E8	SD open file failure
E9	SD Error logging failure
E10	ADC not found

An example of how this will look like in the Particle.io interface here:



Note: the last highlighted message has wrong time stamp because the RTC is missing at this time (E03 means RTC missing).

Type #3: startup message

Every time we reboot the whole system (at startup or if the Arduino is restarting after losing power), the Arduino will send a “wakeup” message to Particle.io.

If we get the following message on the Particle.io, that shows the success of the connection of the cellular module to internet and the communication between the Arduino microcontroller (box “brain”) and the Particle.io cloud system.

Uses:

1. When we are installing sensors, this is the “quick” message that will allow us to verify the sensor box is functioning properly.
2. If we see this message at any other time, it is telling us the box lost power. While it is possible that is just a reality in winter, **at any other time this would probably indicate some kind of malfunction of the battery or the microcontroller.**

Generic format of a message:

BoxIDplusDate Hour:Minute **LTE Setup Done**

where *BoxIDplusDate* is 2-digit BoxID followed by 2-digit year, 2-digit month, and 2-digit date and

where “LTE Setup Done” is text that is always the same (a static wakeup message).

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Example (in screenshot below):

19250610 16:47 LTE Setup Done

In the example:

- Box ID is 19
- Date and time: 2025-06-10 16:47 UTC
- Box is reporting startup and successful connection to LTE network

The screenshot shows the Particle.io dashboard for a device named '19'. The 'General information' section displays the device ID (e00fce6897349dcaadb76012), owner (envensorslab@gmail.com), ICCID (89883070000036202099), IMEI (351457630678285), and serial number (P044AD30952KYV8). The 'State' section shows the device is 'Online', with the last heard time being Jun 10th 2025, 4:47 pm. The 'Location' section indicates that location storage is not enabled. The 'Events' section shows a table of events, with the event 'Environmentdata' highlighted in red. The 'Environmentdata' section shows the event details, including the date and time (Jun 10th 2025, 4:47:54 pm) and the event name (19250610 16:47 LTE Setup Done). The 'Last vitals' section shows various metrics such as cellular signal strength (22%), cellular signal quality (29%), round-trip time (4626ms), RAM used (62.1kB), battery charge (100%), cloud disconnect events (0), and rate-limited publishes (0).

NAME	DATA	DEVICE	PUBLISHED AT
spark/device/diagnostics/update	{\"device\": {\"network\": {\"cellular\": {\"radio_s...	19	6/10/25 at 4:48:00 pm
particle/device/updates/pending	false	19	6/10/25 at 4:47:53 pm
Environmentdata	19250610 16:47 LTE Setup Done	19	6/10/25 at 4:47:54 pm
Setup	Setup Done	19	6/10/25 at 4:47:54 pm
spark/device/test_reset	1580303070	19	6/10/25 at 4:47:54 pm
particle/device/updates/enabled	true	19	6/10/25 at 4:47:54 pm
particle/device/updates/forced	false	19	6/10/25 at 4:47:54 pm
spark/status	online	19	6/10/25 at 4:47:53 pm
spark/device/diagnostics/update	{\"device\": {\"network\": {\"cellular\": {\"radio_s...	19	6/10/25 at 4:45:43 pm

Type #4: system information

Note that the Particle.io system also provides other helpful information about each box, including the signal strength, last connection, etc. **These could be useful for diagnostics, and we may wish to consider pulling these at least once/day with each data message as sensor health checks.**

The screenshot shows the 'Last vitals' section of the Particle.io dashboard. It displays various system metrics for the device '19' as of Jun 10th, 2025, 04:48 PM. The metrics include cellular signal strength (22%), cellular signal quality (29%), round-trip time (4626ms), RAM used (62.1kB of 163.5kB), battery charge (100%), cloud disconnect events (0), and rate-limited publishes (0). A link to 'Download history (docs)' is also provided.